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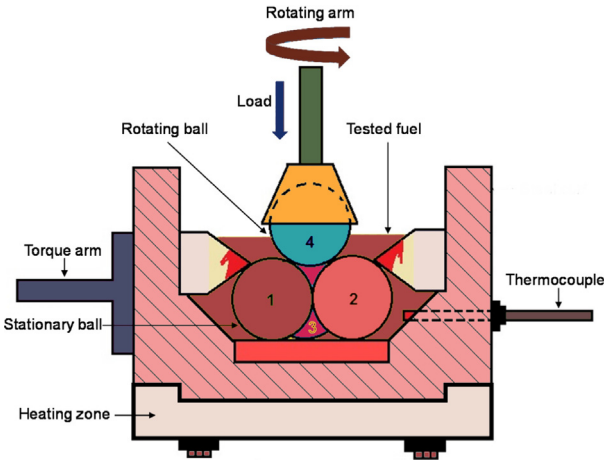
Full Length Article

Development of bio-based lubricant from modified desert date oil (balanites aegyptiaca) with copper nanoparticles addition and their tribological analysis

Yashvir Singh[a,*], Abhishek Sharma[b], Nishant Kumar Singh[c], Wei-Hsin Chen[d]

[a] Department of Mechanical Engineering, Sir Padampat Singhania University, Udaipur, Rajasthan, India
[b] Department of Mechanical Engineering, G L Bajaj Institute of Technology and Management, Greater Noida, UP, India
[c] Department of Mechanical Engineering, Hindustan College of Science and Technology, Mathura, UP, India
[d] Department of Aeronautics and Astronautics, National Cheng Kung University, Tainan, Taiwan, ROC

GRAPHICAL ABSTRACT



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ABSTRACT

Based on the environmental concerns related to the pollution caused by the vehicles and the demand for an alternative to the conventional lubricant resulted in the studies to be conducted that are environmental-friendly. In this study, friction and wear behavior of the surfaces in contact was conducted to observe their tendency during the application of an alternative lubricant to the mineral oil. Desert date oil was having the potential of an alternative to the mineral oil and they were available in abundant amount. During the initial stage, desert date oil was transesterified using the two-step transesterification process and further it was added to the trimethylolpropane. After this process, nanoparticles were added in different amounts to the modified desert date oil. The friction and wear behavior of the modified desert date oil was tested using four-ball tester under different conditions. For the surface topography analysis, SEM and EDS analysis were conducted. The concentration of the nanoparticles added to the biolubricant was considered based on the previous work conducted. The input parameters which were considered during the test consists of normal load, sliding speed, variation in the addition of the nanoparticles to the biolubricant (0.3–1.6% with a gap of 0.3%). Based on the study conducted,

* Corresponding author.
E-mail addresses: yashvir.singh@spsu.ac.in, yashvirsingh21@gmail.com (Y. Singh).

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0.9% concentration of the copper nanoparticles showed a significant improvement in term of reducing COF, wear rate, mean wear scar diameter and improved worn surface morphology was obtained with comparison to the mineral oil. The maximum increment in the properties relative to the tribological analysis was shown when the concentration of the copper nanoparticles to the modified desert date oil increases beyond 0.9% copper nanoparticles during the addition in the amount of 1.3% and 1.6%. Conclusively, the performance of the modified desert date oil with 0.9% copper nanoparticles addition has the potential impact as a lubricant with comparison to the mineral oil and other samples. The modified desert date oil with an application of 0.9% copper nanoparticles could be considered as a suitable alternative to the mineral oil while considering environmental concern and energy saving.

1. Introduction

Around the globe, there is concern relative to the environmental issues especially in the energy and transportation sectors [1]. Among the petroleum products, the use of mineral oil is high due to their application for the lubrication purpose [2]. Many governments are focussing on meeting the higher output demand of the petroleum products and trying to minimize the emission level as low as possible related to both the sectors as discussed by Fontaras et al. [3]. Some environmental issues are there with reference to the usage of mineral oil as they are responsible for the aquatic pollution due to the spillage done after their utilization [4]. By considering these reasons, there is demand for an alternative lubricant which will reduce the cost of the petroleum product and also provide a positive response to the environment during their use [4]. Biolubricant is one of the possibilities which is capable to reduce the usage of the mineral oil. Biolubricants are gaining worldwide popularity and are biodegradable, environmental-friendly, higher viscosity index, etc. [5–7]. Shahabuddin et al. [8] investigated the friction and wear behavior of jatropha oil-based biolubricant at fixed load and sliding speed. The blend of the biolubricant with reference oil was varied in different ratios (10%, 20%, 30%, 40%, and 50%). The 10% blend provides sustainable impact during the friction and wear characterization. There are some of the modifications needed to refine its properties so that it could provide better surface characterization. There are many resources available for the production of the biolubricant including non-edible oils, agriculture residues, animal waste, etc. The utilization of the non-edible oils for the production of biolubricant is one of the convenient processes as they are biodegradable, higher lubricity, high viscosity index, etc. [8–10]. Various sources are available for the availability of the non-edible oils which includes jatropha curcas [11], pongamia pinnata [12], madhuca indica [11], calophyllum inopyllum [13], moringa oleifera [14], schinziophyton rautanenii [15], balanites aegyptiaca [16], etc. Most of the research has been conducted by considering the non-edible oils as mentioned but only a few are related to their application with friction and wear analysis of biolubricant [17]. The research conducted by implementing non-edible oils involved the application of non-edible oils for the production of the biolubricant. Balanites Aegyptiaca is one of the non-edible oils having better suitable properties with reference to their utilization as biolubricant [16]. The purpose of utilization of this oil is relative to the composition of the free fatty acids. Nitiema-Yefanova et al. [18] reported that the amount of linoleic acid is more with respect to azadirachta indica, and jatropha curcas which provides a significant impact during their application to the tribological characterization.

Balanites Aegyptiaca (also known as Desert date) belongs to the family Zygophyllaceae. It could be available in most of the parts of Africa and South Asia in which India is one of the countries. The availability of the desert date oil depends on the climatic conditions and it prefers to grow in most of the arid areas in which the Rajasthan state of India is one of them. So, it could be used as one of the alternatives for the production of the biolubricant due to its abundant availability [19]. A mature desert date tree produces around 11, 000 dry fruits annually consisting of a kernel with 9–14% quantity. The oil produced from these kernels is around 48% as reported in the literature. In Africa, the local community used this plant for the timber and fodder purpose and sometimes fruit is also used for eating purpose. The taste of the fruits is not as good as it contains a high amount of saponins. Around 4,000,000 ton of desert date fruits was obtained which was further processed for the oil extraction process as mentioned in the study conducted by Farag et al. [20]. Based on the previous work mentioned in the literature, important information was obtained with reference to the utilization of Balanites Aegyptiaca oil for the purpose of biolubricant application. None of the studies as mentioned in the literature presented about its utilization as a biolubricant relevant to the application for the transportation sector. Chapagain et al. [16] assess the possibility of the desert date oil suitability for the production of the biodiesel. An amount of 32% to 52% linoleic acid was observed and it consists of a certain amount of fatty acid which is beneficial for its application as a biolubricant.

Some of the properties related to the vegetable oils usage are in a deficient form which needs some improvement. The properties are low oxidation stability, poor fluid flow behavior, low volatility, and reduction in the protective layer thickness due to the presence of double bonds in the carbon chain of the fatty acids which results in more wear [21–24]. To improve these properties, the application of the nanoparticles to the biolubricant is one of the ways as it has self-repair function for the worn areas [25,26]. Zhang et al. [27] stated that the proper dispersion of the nanoparticles to the lubricant interposed a layer between the surfaces in contact. It can be available in many forms like solid, liquid, gas. The nanotechnology is being successfully introduced in many areas, particularly tribology. Nanoparticles have provided a significant results during their addition to the engine oil by reducing the coefficient of friction and wear of the engine parts in contact during motion [28,29] and the mechanism is shown in Fig. 1. Most of the researches were conducted based on the availability of the nano-additives in metallic or metallic oxide forms like Al_2O_3, CuO, Cu, etc [30,31]. Among the nano-additives, copper has gained popularity by providing remarkable results. Padgurskas et al. [32] utilized Fe, Cu, and Co as nano-additives to the lubricant and tribological characterization was performed. The nanoparticles are added as individually and in combination forms. The Cu nanoparticles provide a significant reduction in the wear of the surface in comparison to the other nanoparticles. Based on the nanometer size, particles can easily reach to the contact surface area and forms a protective layer which results in minimum worn surface area. Their shape is likely to display greater rolling effect, advanced elasticity, and better chemical resilience and it will also help in getting better tribological behavior compared to that of only base oil [31,33,34]. The composition, shape, size, and

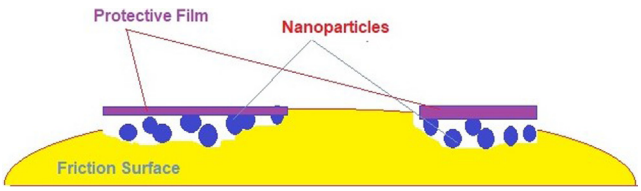


Fig. 1. Formation of the lubricant film with nanoparticles between the surfaces in contact during sliding motion.

concentration are the important properties of the nanoparticles which influences the tribological behavior but the clear description of the concentration influence was not addressed properly [25,35]. In literature, most of the reports stated that the best results are achieved when the concentration is considered below 1% and above 2% [26]. It clearly indicates that there is not an ideal concentration of the nanoparticles. Besides this, many factors are also considered like contact pressure, sliding speed, and the temperature which influences the properties of the base oil. Singh et al. [36] in its previous study examined the tribological behavior of the macadamia oil with the application of copper nanoparticles. The macadamia oil in its present form was used as a reference oil and no process was applied for changing the properties of the macadamia oil. Copper as an nanoparticle additive provides a marginal reduction in the friction coefficient and wear of the contact parts.

The purpose of this work is to investigate the tribological characterization of the mineral oil and copper nanoparticles as an additive to the biolubricant. The biolubricant was developed in a two-step process applied to the desert date oil. Friction and wear test was conducted using four-ball configuration and different conditions (load, temperature, and sliding speed) was considered to check the sustainability of the biolubricant with nanoparticles as additives. This work aims at providing a complete scenery as possible of the nano-additive a usage, depending on the contact pressure, sliding speed, and type of oil.

2. Materials and methods

Fig. 2 shows the methodology adopted for the tribological characterization of the desert date oil. The further details are described in the following sections.

2.1. Development of the chemically modified desert date oil

To make the desert date oil suitable for the tribological application, chemical modification or transesterification process is needed for the improvement of the properties of the oil. The desert date oil was purchased from the local vendor of the Udaipur City, Rajasthan, India. The crude oil purchased contains a higher amount of free fatty acid as reported by Nitiema-Yefanova et al. [18] which needs to reduce below 1%. Two-step acid-base transesterification process was applied consisting of acid and base esterification processes for reducing the free fatty acid amount as shown in Fig. 3. During the acid esterification process, crude desert date oil was mixed with methanol in the ratio of 20:1 in the presence of sulphuric acid in the amount of 0.5% (w/w) as a catalyst. Further, base esterification process was applied to the modified desert date oil obtained during the first step of the reaction. It was mixed with methanol in the ratio of 6:1 by using sodium hydroxide as a catalyst with the presence of 1% (w/w). The desert date methyl ester was produced with the presence of the glycerol which was separated with the help of filter by keeping the solution for a day. The obtained desert date methyl ester was further processed to modify its properties and make it as a suitable biolubricant for the tribological application. The desert date methyl ester was mixed with trimethylolpropane in the presence of 1% sodium methoxide ($NaOCH_3$) for obtaining modified desert date oil as shown in Fig. 4. The chemical reaction for the preparation of the methyl esters was performed in a three-neck flask connected with the condenser as shown in Fig. 5. The obtained oil was procured from the flask as early as possible to prevent the reversible reactions occurs during the process. The three-neck flask was kept in then oil bath while maintaining the temperature around 120 °C. The process was conducted around 8 h to get the desired output. At the last stage, modified desert date oil was separated by performing the filtration process using filter papers. The filter paper consists of very fine mesh due to which the only filtration of the fine particles involved in biodiesel was possible which separated the glycerol from the remaining biodiesel as glycerol is thick [37]. The funnel could only be used for pouring the final product into the flask.

2.2. Nano-lubricant oil preparation

For the further processing of the addition of the nanoparticles to the modified desert date oil, nanoparticles were purchased from the Sisco Research Laboratories Pvt Ltd, India. Five types of samples were prepared for the test in which four samples consists of nanoparticles addition with the modified desert date oil (biolubricant). The nanoparticles were mixed to the biolubricant in the amount of 0.3, 0.9, 1.3 and 1.6% on a weight basis according to the weight of the modified desert date oil. The samples are designated as MO, BA + 0.3%, BA + 0.9%, BA + 1.6% and BA + 1.6% consisting of mineral oil and modified desert date oil with nanoparticles addition in different proportions. The addition of the nanoparticles to the biolubricant was performed using ultrasonication process. The nanoparticles were added to the biolubricant in a flask and sonicated for around 2 h under the influence of ethanol bath at a temperature of 65 °C [36]. The dynamic light scattering technique was used to measure the particles size and the quality of the blending of nanoparticles to the biolubricant. Particles are illuminated by laser to study the speckle pattern. A Malvern Zetasizer Ultra (Model Zen 3600-Red badge) was used with a particle measurement range of 0.3 nm to 1000 nm (diameter).

2.3. Physicochemical properties analysis

Kinematic viscosity, viscosity index, density, flash point, and acid value were measured according to the following ASTM standards: ASTM D445, ASTM 2270, ASTM D93, ASTM D664 respectively. The rheometer was used for measuring the kinematic viscosity at 40 °C and 100 °C temperature. Based on the kinematic viscosity data obtained at particular temperatures, viscosity index was calculated. Density was measured at 15 °C using pycnometer based on weight to the volume ratio. Closed cup tester (Pensky-Martens PMA4) was used for the

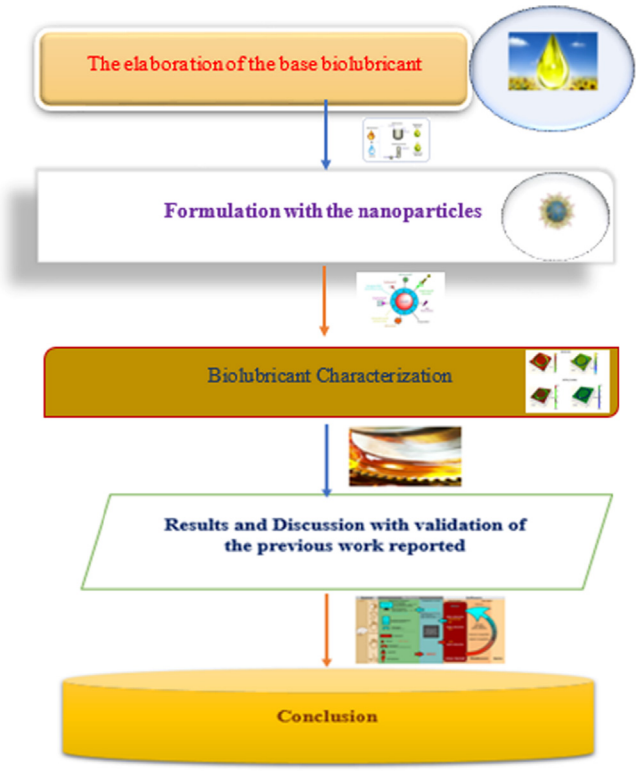


Fig. 2. Schematic image of the methodology adopted for the development and characterization of the desert date oil.

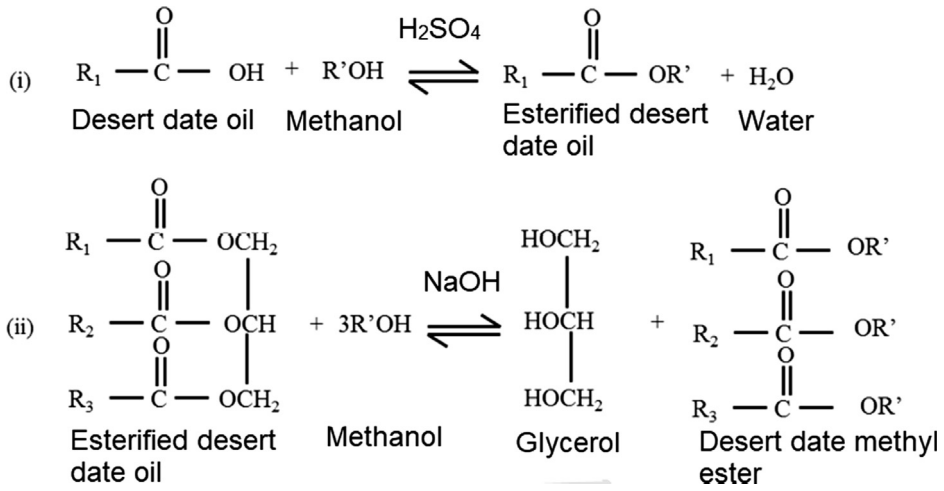


Fig. 3. Transesterification process applied to the desert date oil.

measurement of the flash point. The acid value was calculated through the titration process using KOH as the titration agent. All the tests were performed three times and the average value was reported. Table 1 shows the physicochemical properties of the samples used for the test.

2.4. Friction and wear characterization

2.4.1. Four ball test apparatus

The lubricity of a fluid is an important property which influences the friction and wear behavior of the surfaces in contact. Four ball friction and wear tester were employed for testing the tribological characterization of the different lubricant according to the ASTM D4172 procedure. The testing of the samples for the tribological analysis was performed while keeping the conditions considered during their analysis as per ASTM standards which are globally accepted and considered. The test conditions applied consists of a normal load from 45 N to 95 N load, sliding speed (1.25–3.75 m/s), temperature (75 °C), and operation time (60 min). Fig. 6 shows the schematic image of the four-ball tester employed for friction and wear characterization. For each test, new steel balls (AISI 52100) were used consisting of 13.2 nm mean diameter and hardness in the range 63–65 HRC (Rockwell hardness measured on the C scale). The average surface roughness was 0.3 µm. Fig. 7 shows the surface topography of the steel balls before conducting the test. The energy dispersion X-ray spectrometer (EDS) was utilized for checking the chemical composition of the steel balls which are mentioned in Fig. 8. The loads were chosen based on the highest point for friction that can occur in the engine. This especially occurs during engine start-up at low engine revolutions. Before conducting the experiment for the conditions stated before, the apparatus was cleaned by using the ethanol and further dried using the tissue paper. During the test, 10 ml of the fuel blends were supplied according to the variation in the test performed. After conducting each test, the balls were removed from the apparatus and cleaned with acetone after which tissue paper was used for soaking them which was used for the further surface analysis and wear test. An optical camera was used for the measurement of wear scar diameter and the measurement was done using an image acquisition system mentioning the average length of the horizontal and vertical scars obtained. The average scar diameter was mentioned based on the data obtained after getting a three-time observation of the scar obtained. The average friction torque was obtained using equation (1). For getting the volume wear rate, Eq. (2) was applied.

$$\text{Friction torque, } T = \frac{\mu \times 3w \times r}{\sqrt{6}} \quad (1)$$

where μ = friction coefficient
w = Applied load in N
r = the distance from the centre of the contact surface on the lower balls to the axis of rotation, which is 3.23 mm.

$$\text{Volume wear rate, } k = \frac{\pi}{3t}(R - \sqrt{R^2 - a^2})^2(2R + \sqrt{R^2 - a^2}) \quad (2)$$

where k = the volume wear rate (mm^3/s)
t = the sliding time (s)
R = the radius of the ball (mm)
a = the radius of wear scar diameter (mm)

3. Results

3.1. Physicochemical properties analysis

The results obtained from the physicochemical experiment analysis for modified desert date oil, and nanoparticles addition were presented in Table 1. The density of lubricant varies between 0.92 and 0.9634 g/cm^3. It was observed that the MO provided a slightly higher density

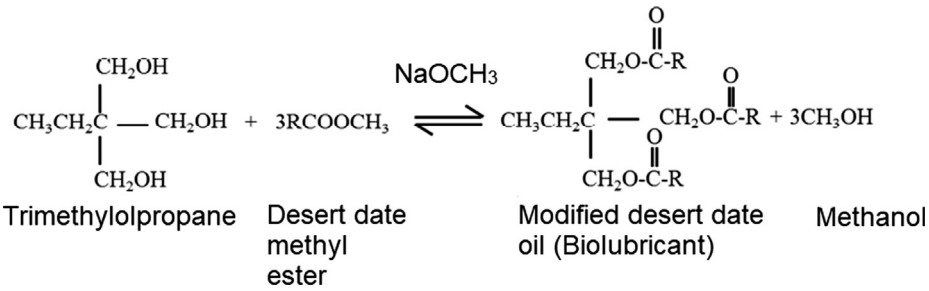


Fig. 4. The process used for the chemical modification of the desert date methyl ester.

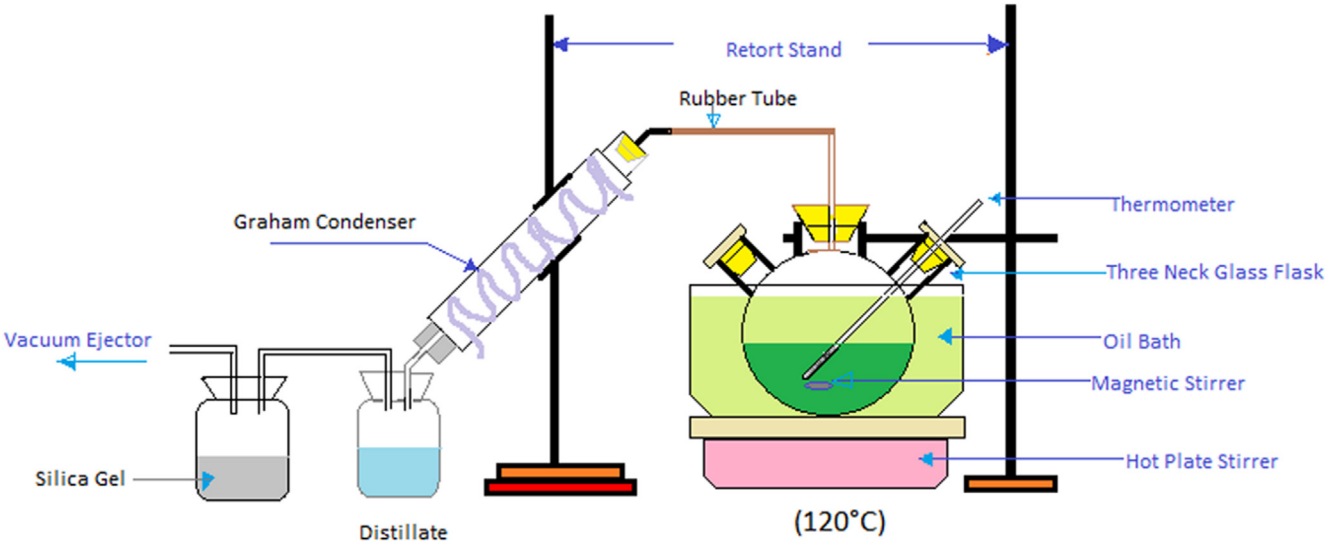


Fig. 5. Experimental set up used for the chemical modification process of the desert date oil.

Table 1
Properties of the samples considered for the tribological analysis.

Samples/Properties	Density @ 15 °C (g/cm^3)	Error (±)	Kinematic viscosity @ 40 °C (mm^2/s)	Error (±)	Kinematic viscosity @ 100 °C (mm^2/s)	Error (±)	Viscosity Index, VI	Error (±)	Flash Point (°C)	Error (±)	Acid value (mg KOH/g)	Error (±)
MO	0.9634	0.0003	18.84	0.32	4.21	0.22	131	0.5	247	0.3	0.53	0.11
BA + 0.3%	0.9215	0.0005	17.42	0.21	4.47	0.19	217	0.6	219.38	0.2	0.32	0.17
BA + 0.6%	0.9219	0.0004	17.68	0.27	4.58	0.15	224	0.3	219.57	1.1	0.30	0.11
BA + 1.3%	0.9324	0.0003	17.75	0.32	4.67	0.19	228	0.2	219.66	1.7	0.35	0.13
BA + 1.6%	0.9363	0.0006	17.89	0.29	4.76	0.17	236	0.3	219.89	0.8	0.38	0.14

(0.9634 g/cm^3) compared to modified desert date oil with nanoparticles addition. The density of modified desert date oil samples increased linearly as a function of the molar ratio and additive concentrations which met the desirable requirement as the density of the lubricant containing hydrocarbon should vary between 0.86 and 0.98 g/cm^3 [38]. It was found that the kinematic viscosities of MO and modified desert date oil with nanoparticles addition varies between 17.42 and 18.84 mm^2/s, respectively at 40 °C. At the operating

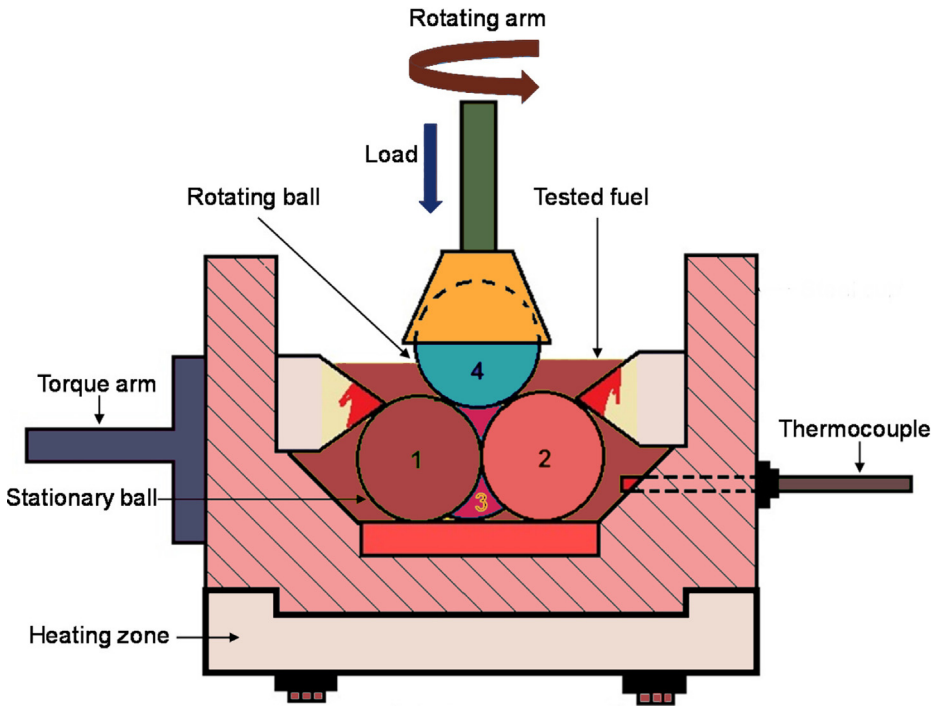


Fig. 6. Schematic image of the four ball tester used for the test.

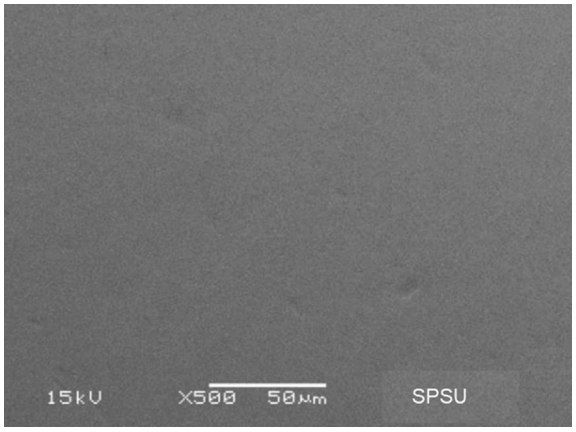


Fig. 7. Surface morphology of the steel balls before the test.

temperature of 40 °C, the production of modified desert date oil with nanoparticles addition recorded low kinematic viscosities. It was noted that the oil became more viscous as to the addition of TMP increased. The high formation of TMP triester is desirable as it has been found to boost the properties of biolubricant in terms of higher viscosity, flash point, and pour point [39]. Moreover, the presence of copper particles in modified desert date oil samples enhanced oil viscosity. Results of the tests showed that oil viscosity increased with the increase in copper nanoparticles concentration. This may due to a high film thickness ratio as the copper concentration increased.

Furthermore, a higher viscosity contributes to a higher VI value. It can be suggested that the higher value of VI of modified desert date oil verified the high formation of TMP ester. The chemical modification caused an improvement in the VI value. A high VI is desirable because it signifies a little change in viscosity over a wide range of temperature. On the contrary, a low VI indicates a large change in viscosity over a wide range of temperature, inferring that the lubrication film becomes extremely thin at high temperatures and extremely thick at low temperatures [38]. The formation of long molecular chain and mass of TMP ester contributes in high VI [40]. The presence of nanoparticles in modified desert date oil samples significantly improved the VI compared to MO. The VI significantly increased as a function of copper nanoparticles concentration ratio. Flash point shows the lowest temperature at which it can vaporize to form an ignitable mixture in air [38,39]. MO recorded the highest flash point of 247 °C followed by the BA + 1.6% (219.89 °C). In this study, it was noted that the flash point of modified desert date oil samples was much lower than those of MO. This was due to the existence of methyl esters in the modified desert date oil samples during the reaction between modified desert date oil esters and TMP in developing TMP ester. Prior to this reaction, the transesterification process to produce the methyl esters which reduced the properties of the crude oil. The flash point value was reduced due to the presence of unreacted methyl esters. The presence of nanoparticles particles in modified desert date oil increased the flash point by 1.5%.

Acid value also contributed to the quality of the lubricant feedstock. The results obtained from the study depicted that the acid value of the modified desert date oil samples with nanoparticles was fairly lower than those of MO. A lower acid value contributes to higher oil quality. Evidently, the presence of copper nanoparticles samples increased acid values. However, only a slight improvement was found with the increment of less than 0.6% copper nanoparticles addition. It was noted that the modified desert date oil portrayed physicochemical properties comparable with those of MO, in terms of high VI, lower acid value and density.

3.2. Coefficient of friction analysis

Figs. 9 and 10 show the effect of the addition of the nanoparticles to the fuels tested at different sliding speed and load. The load was applied from 45 N to 95 N which has been stated in the previous sections. The applied loads and sliding speeds are having a significant contribution during the analysis of the fuel samples characterization. With an increase in sliding speed, there is a reduction pattern in the coefficient of friction amount observed when 45 N load applied. The COF was

Elements	Wt (%)	Elements	Wt (%)
Fe	96.72	Mn	0.33
C	0.89	S	0.03
Cr	1.7	P	0.02
Si	0.28		

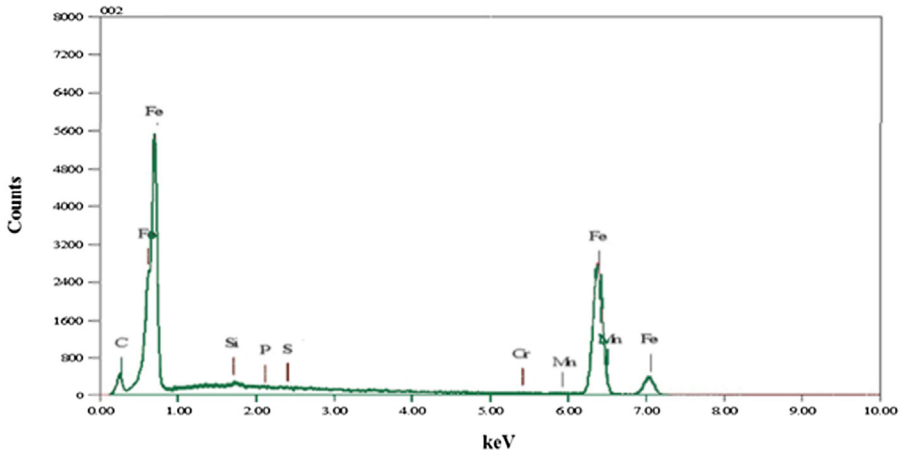


Fig. 8. The EDS spectrum obtained for the steel ball surface.

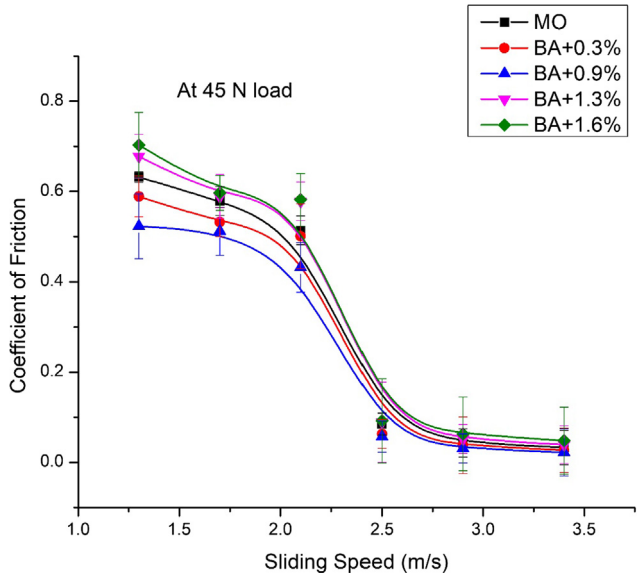


Fig. 9. Coefficient of friction of the samples tested at 45 N normal load.

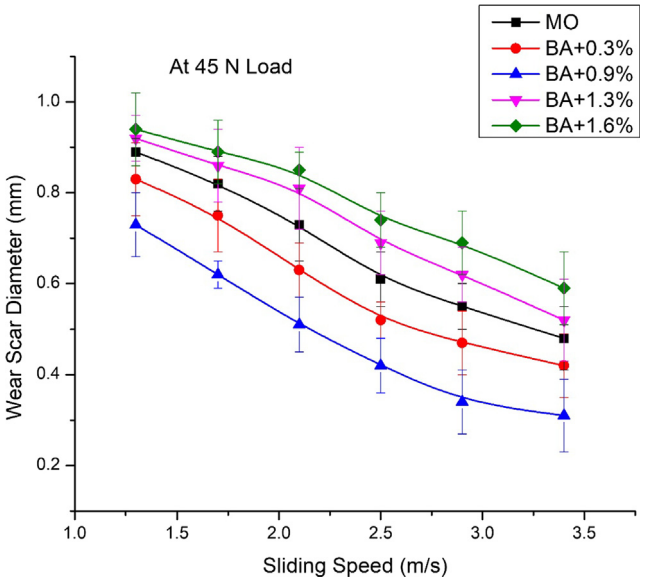


Fig. 11. Wear scar diameter observed for all samples at 45 N load.

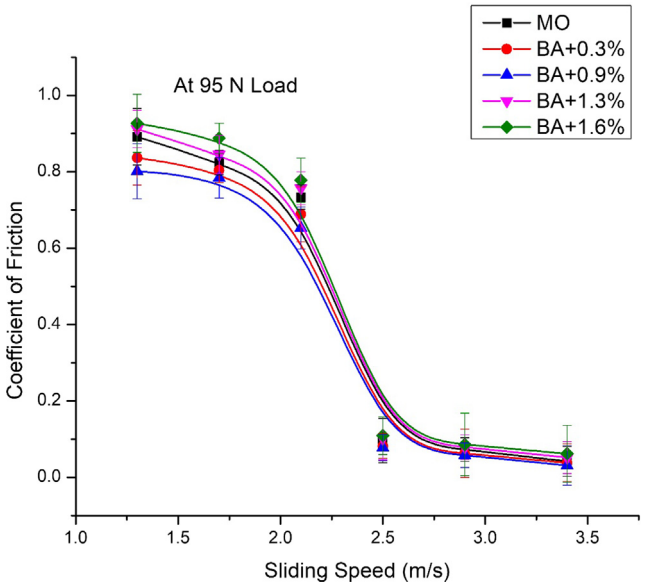


Fig. 10. Coefficient of friction with different sliding speed for all the samples at 95 N load.

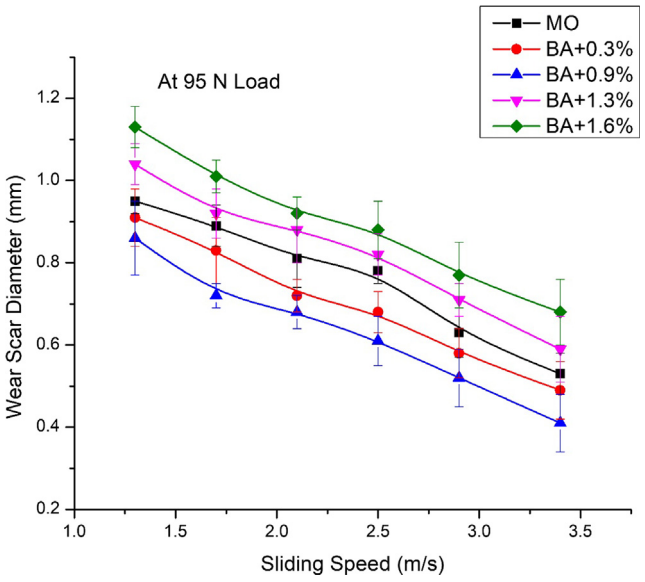


Fig. 12. Wear scar diameter for all the samples at 95 N load.

obtained less with the application of biolubricant containing 0.9% copper nanoparticles addition. When the amount in the addition of the nanoparticles increased copper nanoparticles and beyond the 0.9% copper nanoparticles concentration to the biolubricant, increment in the amount of COF was observed. The obtained quantity of COF was more in comparison to the mineral oil with respect to the increment in the quantity of nanoparticles beyond 0.9% copper nanoparticles. The same pattern has been observed with an increase in the applied load consisting of 95 N. There was around 16 to 21% increment in the value of COF was obtained during the application of 95 N load.

3.3. Wear scar diameter

Figs. 11 and 12 show the mean wear scar diameter obtained for the samples obtained at normal load of 45 N and 95 N respectively with respect to the sliding speed. The wear scar diameter was observed with the help of a microscope equipped with a view 7 software which is capable of measuring them in different units. More wear scar diameter was obtained at higher load. The increase in the amount of applied pressure on the surface results in more scar. With respect to the sliding speed, decrease in the value of mean wear scar diameter was observed for both the normal load of 45 N and 95 N. Around 5% difference in the data was observed for the applied loads. There is also a significant impact of the parameters considering load and sliding speed which has also been stated in the work conducted by Guzman Borda et al. [41].

The addition of the copper nanoparticles to the biolubricant played a significant role in a decrement of the wear scar diameter. Up to the addition of the 0.9% copper nanoparticles to the biolubricant, decrease in the value of the mean wear scar diameter was observed. Around 8% decrement was obtained in comparison to the mineral oil. After further addition of the nanoparticles to the biolubricant, an increase in the value of mean wear scar diameter was observed and the difference was around 7%. The copper nanoparticles in the form of solid lubricant maintained a strong protective film between the surfaces during the sliding motion. The copper nanoparticles deposited on the friction surface and performed tribochemical reactions which contributed towards the anti-wear characteristics [32]. The chemical reaction occurs

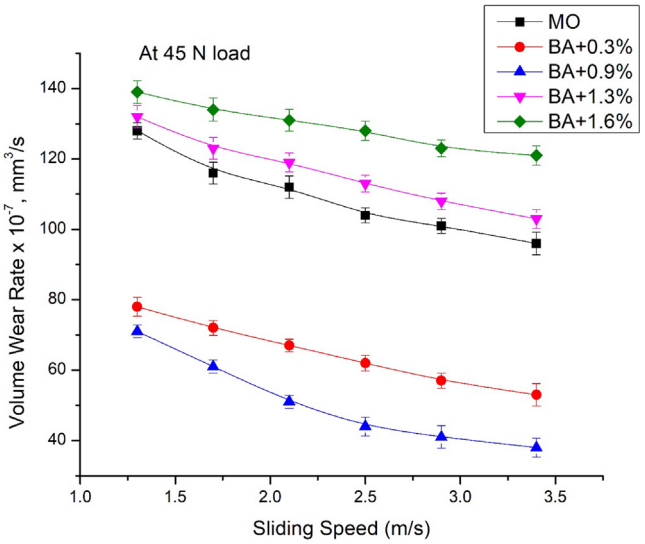


Fig. 13. Volume wear rate for the samples at 45 N load.

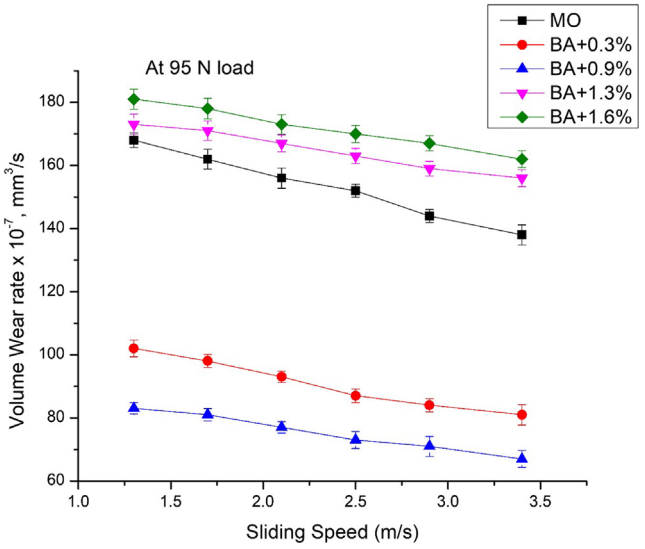


Fig. 14. Volume wear rate for the samples at 95 N load.

between the additives functional groups and metal atoms present on the surface which further attributed towards the formation of a protective lubricant film [42].

Friction and wear variations of suspensions as a function of nanoparticles content are related to size, hardness and deposition of nanoparticles on wear surfaces. Friction coefficient values in all tests are for a mixed lubrication regime. According to Chinas and Spikes' mechanical entrapment theory, nanoparticles penetrate in the contact area and then deposit on it because they are smaller or similar in size to lubricant film thickness [43]. On the other hand, nanoparticles have a deleterious effect in some cases, increasing either friction or wear [44,45].

After an optimum concentration of the nanoparticles, a boost in the value of wear scar diameter was observed which was due to the ploughing effect resulting in deep and uneven groves on the surface. With an increase in the copper nanoparticles concentration on the asperity valleys formed on the worn surfaces results in higher stress concentration and generates more amount of force. There is a decrease in the movement between the nanoparticles on the surface which contributed to more amount in the value of wear scar diameter. Finally, it has been revealed that the concentration of the nanoparticles up to a limit of 0.9% of the copper results in the formation of a better protective film as they are capable to withstand the tribochemical reactions occur on the surface.

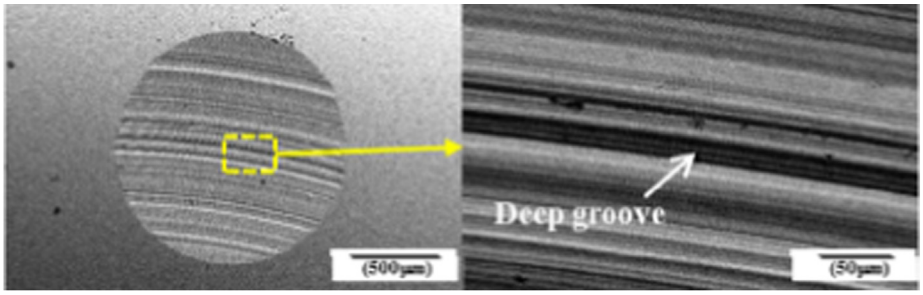
3.4. Wear rate

Figs. 13 and 14 show the wear rate of the samples tested at different load and the graphs are plotted by considering the sliding speed as the base axis. More amount of the wear rate in volume was observed at 95 N load with comparison to the 45 N load. The same trend with respect to the sliding speed was also observed with respect to the samples considered for the test. The effect of sliding speed and load while considering them as parameters are the same and the reason behind the changes in their values are equivalent as stated before. The reasons for their changes are also in accordance with the previous work conducted by the author [36,46]. The chemical modified desert data oil also plays a contribution to the wear rate. The modified desert date oil consists of a higher viscous oil with more amount of TMP ester which results in the formation of a protective film between the surfaces in contact. Singh et al. [4] mentioned the details about the reduction in friction and wear amount when castor oil was utilized on the steel surface. The adsorption of the acid molecules of castor oil on the steel surface results in the formation of a protective layer which prevents friction and wear of the surface.

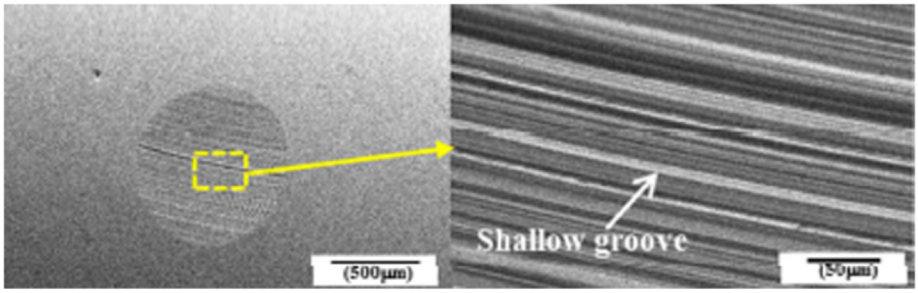
The addition of 0.9% copper nanoparticles to the biolubricant improves the reduction in the amount of wear rate with respect to the mineral oil. The presence of the copper nanoparticles improves the decrement in the amount of COF and mean wear scar diameter which also decreases the amount of material removal from the surface by forming a better protective film. The results are in accordance with the work conducted by the Singh et al. [36] in which the lower concentration of the copper nanoparticles improved the tribological behaviour of the base oil considered for the study. The increase in the amount of nanoparticles beyond a certain limit results in an increase in volume wear rate due to the agglomeration of the nanoparticles on the pits and valleys formed on the worn surface. The 0.9% copper nanoparticles addition to the biolubricant showed an improved behaviour in terms of volume wear rate obtained at different conditions.

3.5. Worn surface characterization

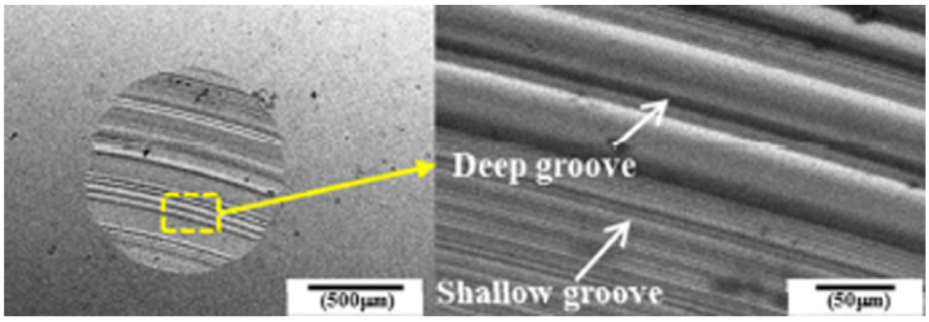
Figs. 15 and 16 show the worn surface topography obtained for the samples and their EDS spectrum. The surface morphology and EDS spectrum as shown in Figs. 15 and 16 are obtained when a higher load of 95 N and lower sliding speed was considered. Fig. 15(a) shows the worn surfaces observed during the use of mineral oil. Deeper parallel grooves are formed on the surface. Fig. 16(a) shows the EDS spectrum containing around 5.9% of C element on the worn surfaces. Abrasive wear was occurred on the surface of the balls due to the polished wear mechanism resulting in the breakdown of the lubricant film. The lubricant film formed by the mineral oil does not support for the greater duration when the load was applied. Fig. 15(b) shows the worn surface when 0.9% nanoparticles are added to the biolubricant during which smoothest surface finish was obtained with comparison to the mineral oil. Few parallel lighter grooves are formed on the surface. Fig. 16(b) shows the EDS spectrum in which additional elements of copper (C = 1.23 wt%) was obtained on the worn surface when lubricated with BA + 0.9%. This indicated the deposition of Copper nanoparticles on the worn surface during sliding contact. With this evidence, it can be justified that BA + 0.9% has the ability to form a protective lubricant film which remains unbroken for the longer during of the sliding contact between the surfaces. The wear of the surfaces increases when nanoparticles concentration increases as shown in Fig. 15(c) and (d). Uneven and deep grooves get formed on the worn surface when nanoparticles concentration increased to 1.3% and 1.6% The particles gets cleaved on the valleys formed on the surface which results in greater stress concentration on the surfaces in contact. The particles got



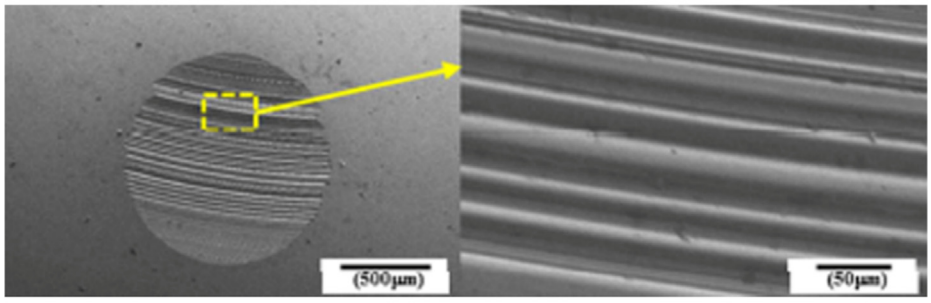
(a)



(b)



(c)

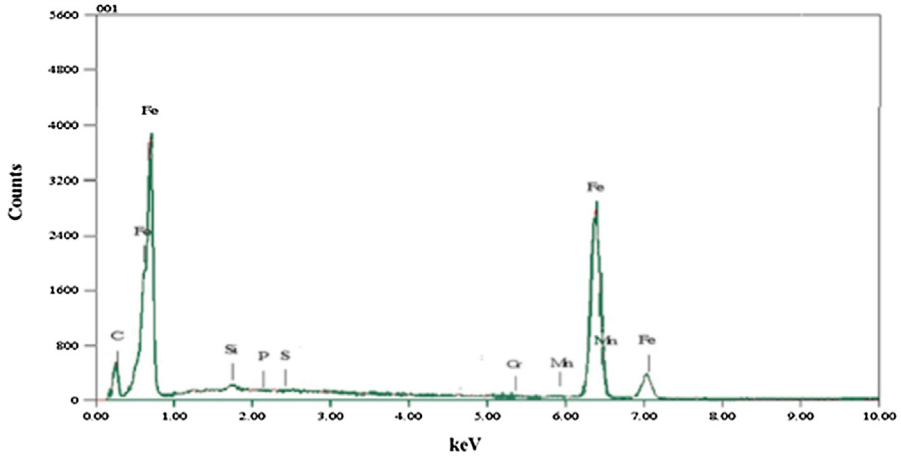


(d)

Fig. 15. SEM images of the different samples (a) Mineral oil (b) BA + 0.09% (c) BA + 1.3% (d) BA + 1.6%.

entrapped into the valley and prevent further motion between the surfaces resulting in higher stress concentration. This phenomenon results in the abrasive wear of the surface and increases COF, mean wear scar diameter and wear rate. This was also supported by the study of Golshokouh et al. [47] which stated that the excess wear occurs due to the abrasiveness of the surface which was due to the deep ploughing on the worn surface. These results revealed that the more viscous modified oil with a minimum concentration of nanoparticles is suitable for providing a better lubricant film between surfaces during sliding contact.

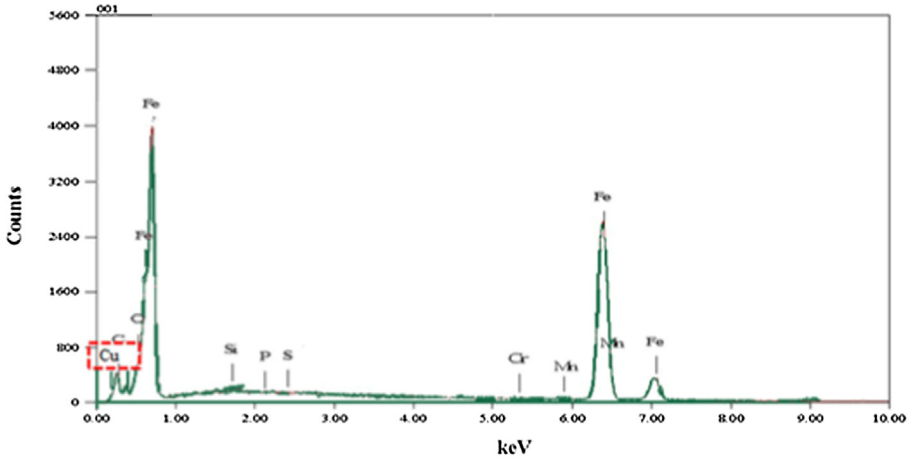
Elements	Wt (%)	Elements	Wt (%)
C	5.9	Mn	0.4
Fe	92.78	S	0.04
Si	0.19	P	0.06
Cr	0.05		



(a)

Fig. 16a. The EDS spectrum of the mineral oil used for the test.

Elements	Wt (%)	Elements	Wt (%)
C	7.12	Mn	0.25
Fe	90.12	S	0.02
Si	0.28	P	0.03
Cr	0.26	Cu	1.23



(b)

Fig. 16b. The EDS spectrum of the biolubricant with 0.9% copper nanoparticles concentration.

4. Conclusions

Based on the experimental conditions by considering nanoparticles as additives, an enhancement in the tribological characteristics of the biolubricant are achieved. By mixing nanoparticles up to 0.9% copper concentration, suitable boundary lubricant film was formed between the surfaces in contact which are capable to withstand the applied loads considered during the test. The following conclusions are obtained from this work:

1. Maximum COF, wear and wear scar diameter occurs at 1.25 m/s minimum sliding speed and a maximum load of 95 N while adding copper nanoparticles up to 0.9% in comparison to the mineral oil. Better results in terms of tribological characteristics are obtained up to a limit of 0.9% copper nanoparticles addition.
2. Better physicochemical properties of the modified desert date oil with copper nanoparticles up to 0.9% was obtained in terms of viscosity index, density, and acid value. It could be suggested that a high viscosity of the biolubricant is capable to form a protective lubricant film between the surfaces in contact.
3. The tribological properties of the modified desert date oil get worsened when the concentration of the nanoparticles increases up to 1.6%. Increase in the amount of copper nanoparticles after a certain limit didn't provide better characteristics during its analysis.
4. Based on EDS and SEM analysis, smooth surface was obtained up to the addition of 0.9% copper nanoparticles addition with comparison to the mineral oil.
5. Based on overall results, the addition of nanoparticles to a limit of 0.9% copper nanoparticles are capable to provide an improvement in the tribological performance of the biolubricant. The applications of these type of biolubricant to the transport sector contribute to pollution reduction and energy saving.

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